

Stefano Biondi

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EDUCATION

Ph.D. in Civil and Coastal Engineering,	University of Florida
M.S. in Civil Engineering, Specialization - <i>Hydraulics and Hydrodynamics</i>	University of Padova
B.S. in Civil Engineering,	University of Padova

EXPERIENCE

Research Coastal Engineer, NIST, Gaithersburg, MD Analysis of wave fields generated by hurricanes	Jul 2025 - current
Experimental Researcher, ERDC, Vicksburg MS Setting up and performing fluid dynamics experiments	May 2025 - Jul 2025
Field Data Analyst, GTM Research Reserve, FL Data collection and analysis in the Florida coast.	Oct 2023 - Oct 2024
Physical Modeler, University of Padova, Italy Wave flume physical experiments to quantify wave energy dissipation	Apr 2023 - Jun 2023
Research Scholar, University of Florida Numerical research on HPC techniques for high resolution dynamic models	Jul 2022 - Aug 2023
Calculus/Physics Tutor, University of Padova, Italy Tutored mechanics, physics, and calculus to first-year college students.	Oct 2019 - Nov 2021

CONFERENCES

ICCE, Galveston, TX Presenting: Numerical Analysis of Storm Wave Forces and Attenuation in the Built Environment Co-authoring: Validation of wave uplift design equations for elevated structures	May 2026
OSM, Glasgow, Scotland Co-authoring: Quantifying Sediment Transport and Morphological Changes around Oyster Reef Breakwaters: Insights from Large-Scale Experiments	Feb 2026
AGU, New Orleans, MS Presenting: Numerical investigation of wave forces in the built environment	Dec 2025
Hipergator Symposium, University of Florida Lightning talk on HPC and CFD challenges in wave modeling	Oct 2024
WolfDynamics OpenFOAM Workshop, Genova, Italy Presenting: Quantifying wave energy dissipation by wavescreens	Jul 2023
COP28, Dubai, UAE panelist: Access to Supercomputing in Climate Research	Dec 2023

PUBLICATIONS

S. Biondi & D. Slinn, *Wave Forces and Attenuation in Coastal Building Arrays: A Parametric Numerical Study*, Coastal Engineering, *preprint* DOI: [10.2139/ssrn.6955358](https://doi.org/10.2139/ssrn.6955358) .

S. Biondi, C. Favaretto, A. Canestrelli, L. Martinelli, P. Ruol, *Investigation of wave interaction with partially submerged wave screens*, Applied Ocean Research DOI:[10.1016/j.apor.2026.105100](https://doi.org/10.1016/j.apor.2026.105100)

S. Biondi, Pinton, D., Chu, C. C., Dix, N., Smyth, A., Reisinger, A. J., & Canestrelli, A. (2026). Observed nitrogen accumulation during salinity and temperature surges in a coastal dam controlled lake. *Journal of Environmental Management*, 405, 129590. DOI: [10.1016/j.jenvman.2026.129590](https://doi.org/10.1016/j.jenvman.2026.129590)

S. Biondi, D. Pinton, C. Chu, N. Dix, A. Canestrelli, *Quantifying Constituent Discharge through a Bi-directional Estuarine Dam Using Water Level Observations*, Regional Studies in Marine Science, (submitted - under review).

S. Biondi, A. Canestrelli, L. Martinelli, P. Ruol, *Quantifying wave energy dissipation by wavescreens with high resolution 3D numerical modeling*. Master thesis, 2023. DOI: [20.500.12608/46031](https://doi.org/20.500.12608/46031)

H. Fischmann, S. Williams, J. Crabill, N. Dix, **S. Biondi**, D. Pinton, A. Canestrelli, K. Todd-brown, J. K. Reimer, J. Dacey, A.J. Reisinger, K. Perez, S. Baker, K. Dietz, V. Powell, J. Kindelspire, A. Smyth, C. Angelini, *Strategic Management of multiple bivalve species across an estuarine gradient to counteract eutrophication*, Estuaries and Coasts (submitted -under review)

S. Biondi, *Dissolved Oxygen Metric Choice Changes Estuarine Water Quality Assessment*, Environmental Monitoring and Assessment, *preprint* DOI: [10.21203/rs.3.rs-9890420/v1](https://doi.org/10.21203/rs.3.rs-9890420/v1)

S. Biondi, *Source-Aware Screening of Long-Term Salinity Trends in Public Estuarine and Coastal Monitoring Records*, Estuarine, Coastal and Shelf Sciences, *preprint* DOI: [j.ecss.2026.110103](https://doi.org/j.ecss.2026.110103)

S. Biondi, *A 52-Event Field Assessment of Cross-Shore Wave Transformation in 1D XBeach Non-Hydrostatic at a Barred Beach (Duck, NC)*, Ocean Engineering, *preprint* DOI: [10.2139/ssrn.6996528](https://doi.org/10.2139/ssrn.6996528)

S. Biondi, *Pycoastal: a python tool for coastal engineering*, 2025. DOI: [10.13140/RG.2.2.13257.40806](https://doi.org/10.13140/RG.2.2.13257.40806);

AWARDS

Outstanding Poster, STEM category, Sigma Xi chapter
CERF 2025 Student Award, Richmond, Virginia

SKILLS

Coding: Python, C++, Fortran, MATLAB, Bash, SLURM, HTML

Field and Lab Sensors: ADCP, ADV, HYCAT, EXO2, ProDSS, HOBO

Models and Tools: OpenFOAM, Delft3D, HEC-RAS, SWAN, HPC, XBeach, FLOW-3D, ADCIRC

MENTORING

Graduate Peer Mentor, University of Florida

2024 – 2025

Mentored junior graduate students on numerical modeling workflows and HPC job submission.

Pietro Lazzarini - *Quantifying morphological changes driven by oyster reef breakwaters under different tidal and wave conditions* DOI: [20.500.12608/74350](https://doi.org/20.500.12608/74350)

Federico Smaniotto - *Where Do Oyster Larvae Go? Analyzing Connectivity and Restoration Strategies*

Through Field-Calibrated Larval Transport Modeling in the Guana-Tolomato-Matanzas Estuary, Florida
DOI: [20.500.12608/85325](https://doi.org/10.500.12608/85325)

EXTRACURRICULAR

Gator Motorsports Aerodynamics Team, UF: Chassis design for FSAE competition.

Swimming: 7x Italian National Finalist, Regional Record Holder (50 m Backstroke).

Masterchef: Finalist of the 2018 Italian edition.

LANGUAGES

Italian, English (*Native*), Spanish (*Intermediate*), French (*Basic*).